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IoT Industry Applications and Solutions

With the development of the Internet of Things (IoT), its impact has penetrated into every aspect of society. IoT is everywhere, from smart homes to smart cities.

Here, we analyze existing problems in four IoT industry scenarios: smart city, all-in-one smart home, Internet of Vehicles (IoV), and industrial IoT. We also present solutions provided by IoT and success stories.

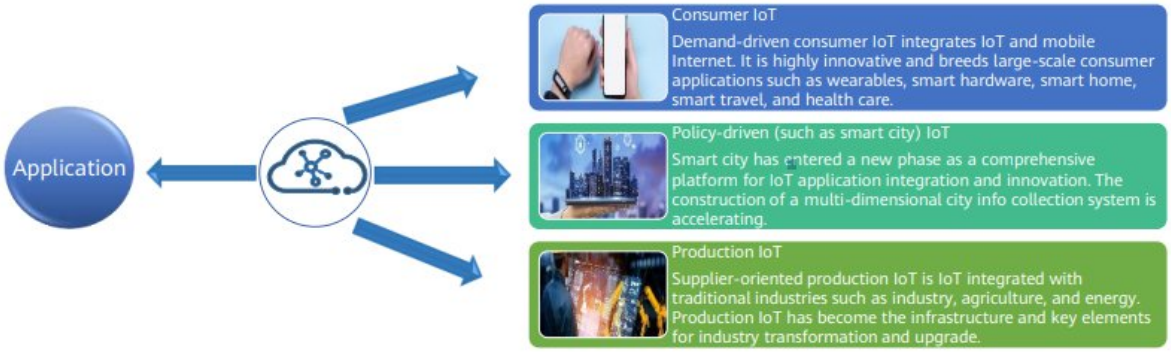
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1. IoT Application Types

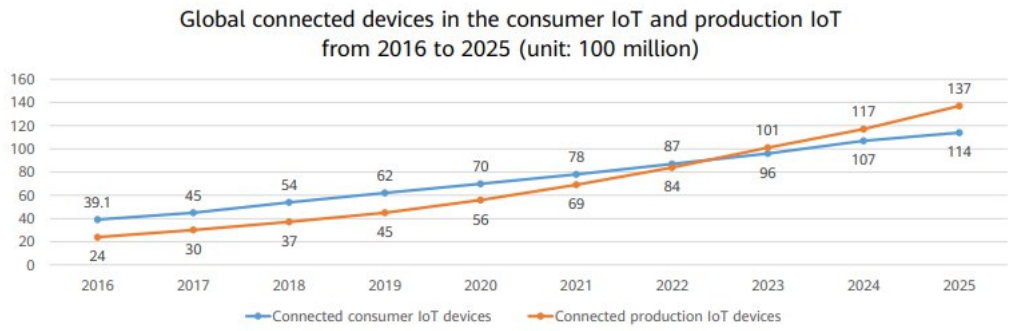
The Three Main Types of IoT Applications

Consumer IoT, policy-driven (such as smart city) IoT, and production IoT. Together, policy-driven IoT and production IoT are called industry IoT.



Global Perspective: Surging Production IoT Connections

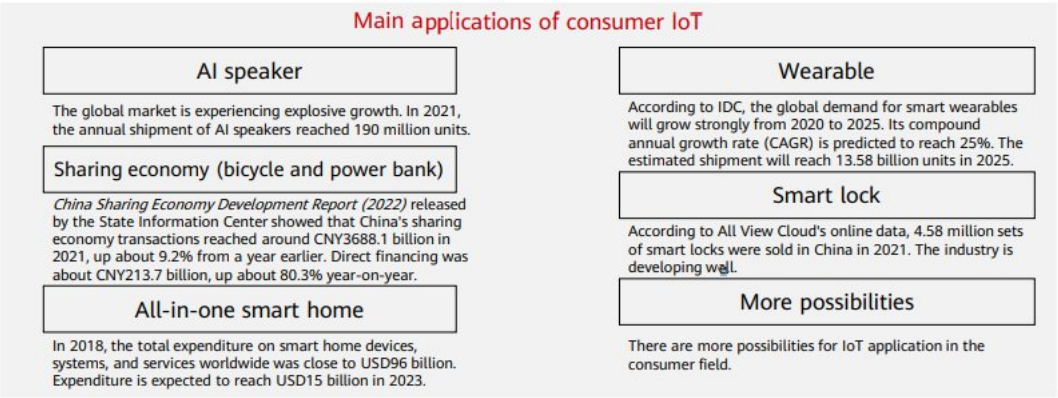
Production IoT and consumer IoT are developing at the same time. Both parties want to solve and optimize development issues of the industry, energy, and transportation sectors, as well as enterprises.



The adoption of IoT in more industries brings more device connections. Production IoT is becoming the promising player instead of consumer IoT. GSMA Intelligence, one of the three top international organizations in mobile communications, predicts that the number of production IoT connections will increase by 4.7 times from 2017 to 2025, while that of consumer IoT connections will increase by 2.5 times.

Consumer IoT

Internet enterprises optimize this ecosystem and innovation for consumer products, driving the expansion of the industry.

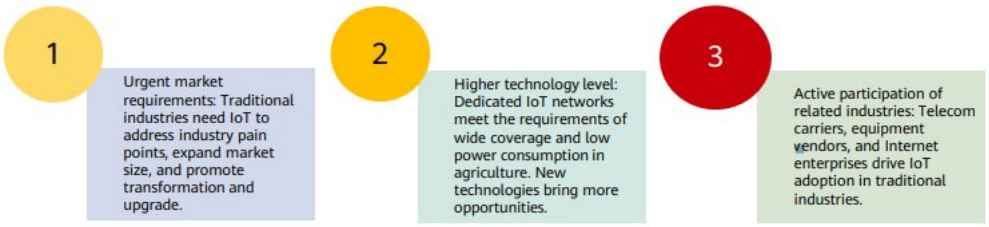


Policy-Driven IoT Example: Smart Cities

Led by the concept of "digital twin cities", urban IoT will have a bigger scale, wider applications, and higher integration. However, city pain points, requirements, and deployment costs affect the development of smart city IoT.

No.	Industry	IoT Application
1	Safety	With policies such as safety and Sharp Eyes projects, the security industry is witnessing rapid growth. In 2017, the security market scale in China exceeded CNY630 billion, and the number of security manufacturers exceeded 7000. Large expansion of the security industry is a favorable environment for IoT. IoT integrates with smart security more deeply, and the number of connected smart security devices is surging. "AI + security" has become a typical feature of the IoT application in the security industry.
2	Public utilities	Public utilities use low power wide area networks (LPWANs) for intelligent upgrade. Intelligent upgrade of public utilities such as urban water supply, power supply, and heating is the most typical livelihood project of smart city in recent two years. More suitable LPWANs such as NB-IoT and LoRa are introduced into public utilities.
3	Firefighting	A survey on major firefighting enterprises in the past two years showed that overall revenue growth rate remained between 20% and 40%. Only a few managed to reach 100%. According to a conservative estimate that the CAGR of China's smart firefighting industry will reach 20% in the next five years, the market scale of China's smart firefighting industry will reach about CNY100 billion in 2026.
4	Electric bicycle	*China Sharing Economy Development Report (2022)* released by the State Information Center showed that China's sharing economy transactions reached around CNY3688.1 billion in 2021, up about 9.2% from a year earlier. Direct financing was about CNY213.7 billion, up about 80.3% year-on-year. The successful pilot of the electric bicycle management solution will witness large-scale connections of electric bicycles in smart cities.

Production IoT Applications Are New Opportunities



Main Applications of Production IoT

No.	Domain	IoT Application
1	Industrial IoT	According to MarketsandMarkets, infrastructure and industrial development in emerging economies such as China and India have promoted the growth of industrial IoT in Asia Pacific. The development mode of the industrial IoT has four application modes: intelligent production, network-based collaboration, personalized customization, and service-oriented transformation.
2	Agricultural IoT	According to the *14th Five-Year Plan for National Informatization*, smart agriculture has evolved, the informatization rate of agricultural production has reached 27%, and the annual online retail sales of agricultural products have exceeded CNY800 billion. 100 national digital agricultural innovation bases have been built, and 200 agricultural and rural informatization demonstration sites have been certified. The agricultural and rural big data system has launched, and a national agricultural and rural big data platform built, forming a map of agricultural and rural data resources. The innovation system of agricultural and rural informatization is further improved. The independent innovation capability continues to grow. New breakthroughs have been made in key core technologies and products. More than 60 national digital agricultural and rural innovation centers, branches, and key labs have been built.

Smart city

- For IoT to flourish, three conditions must be met: a dense population, a strong foundation of industrialisation, and a nationally integrated market.
- Only China fulfils all three conditions.
- In November 2008, IBM proposed the concept of a smarter planet and released the plan in China in August 2009. They officially launched the smarter planet strategy in China.
- Smart cities are pilot projects under the concept of a smarter planet. Smart cities face many difficulties and challenges.
- The section explains the difficulties faced by smart cities in various scenarios and finds suitable solutions

Common problems

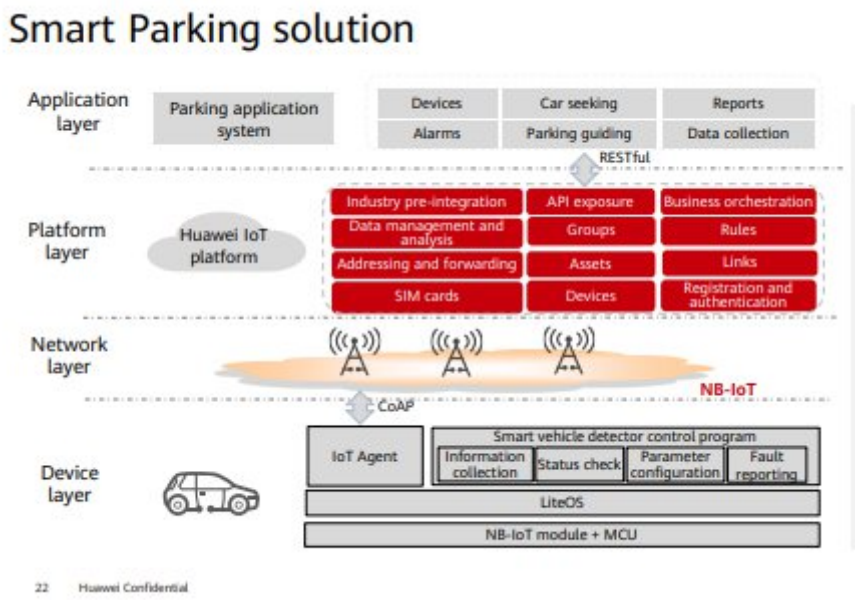
Parking difficulties are a common problem in large and medium-sized cities in China. The average parking time for each vehicle is 18 minutes. Parking difficulties lead to parking violations and traffic congestion.

The vacancy rate of parking spaces in some cities such as Beijing, Shanghai, Guangzhou, and Shenzhen is quite high, reaching 44.6%. The vacancy rate is especially high in underground car parks.

The lack of real-time information exchange between car parks and drivers is a problem. This results in low utilisation of parking spaces. The use of parking spaces in residential areas during the day is low, while almost all parking spaces in office areas are available at night.

This problem causes drivers to constantly search for parking spaces, increasing traffic congestion, pollution emissions, and reducing the efficient use of parking spaces.

Parking problem solution



The Smart Parking solution consists of several layers:

Platform Layer: This layer uses an IoT platform to provide smart parking applications with basic capabilities in connectivity management, data management, and device management. The platform supports flexible and rapid deployment through open and standardised APIs.

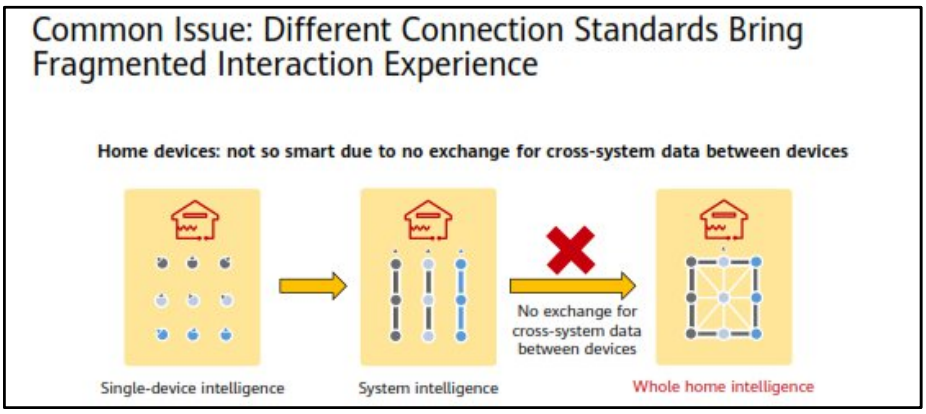
Network Layer: This layer uses NB-IoT technology that has a wide coverage, multiple connections, and simplified architecture. It meets discrete parking requirements to reduce operator installation and maintenance costs.

Device Layer: This layer uses an integrated NB-IoT module that enables data parsing on the device to hide access differences. This layer also includes programme control of smart vehicle detectors, collection of IoT Agent status information, parameter configuration, and crash reporting.

Application Layer: This layer includes a smart parking application system for data management and analysis. This layer also involves business settings, group rules, industry pre-integration API exposure, as well as asset, link, and SIM card management.

With this Smart Parking solution in place, users can leverage IoT technology to optimise parking space usage, reduce time spent searching for parking spaces, and improve operational efficiency.

3. All-in-One Smart Home Solution

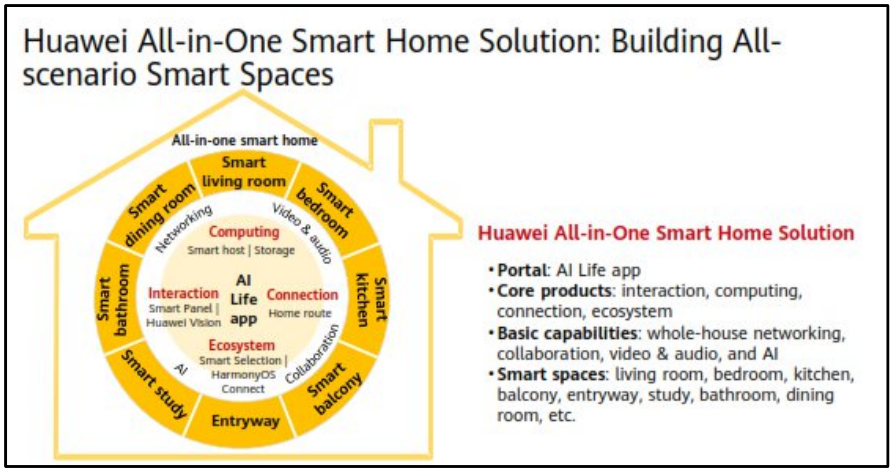


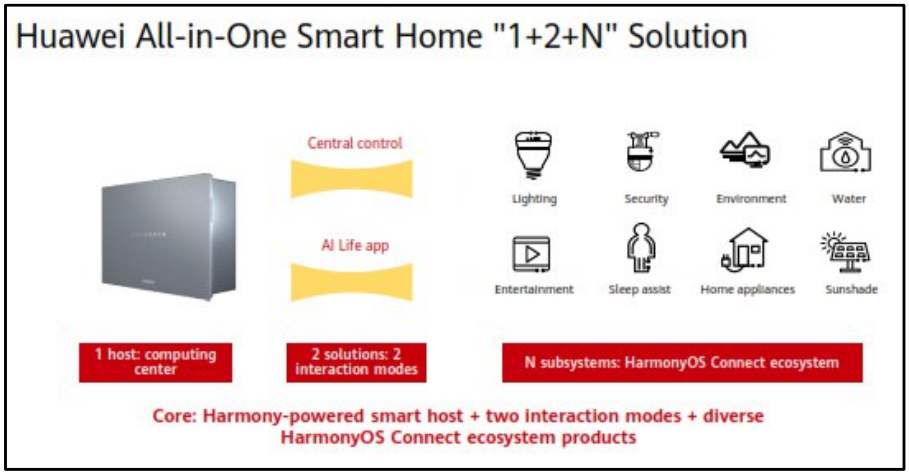
Smart home industry: Different connection standards bring fragmented interaction experience.

The statement indicates that different connection standards in this industry result in a fragmented interaction experience. "Fragmentation" here refers to the division or fragmentation that occurs in the context of using various connection standards in the smart home industry, which can then lead to an interaction experience that is not uniform or integrated between devices. This ultimately can cause incompatibility between devices, user discomfort, resource wastage, and functional limitations.

In other words, the lack of ability or facilities to transfer and share information between devices renders the system considered not very smart. In the context of smart homes or connected technology, "cross-system data exchange" is crucial to creating an integrated and intelligent experience.

"The Huawei All-in-One Smart Home Solution" is a comprehensive solution offered by Huawei for smart homes that encompasses various aspects (Bring digital to every home/hotel)





"Huawei All-in-One Smart Home '1+2+N' Solution" is a comprehensive smart home concept by Huawei. It revolves around three key components:

1 Host: Computing Center

A central computing unit that acts as the core of the smart home, managing and controlling all connected devices.

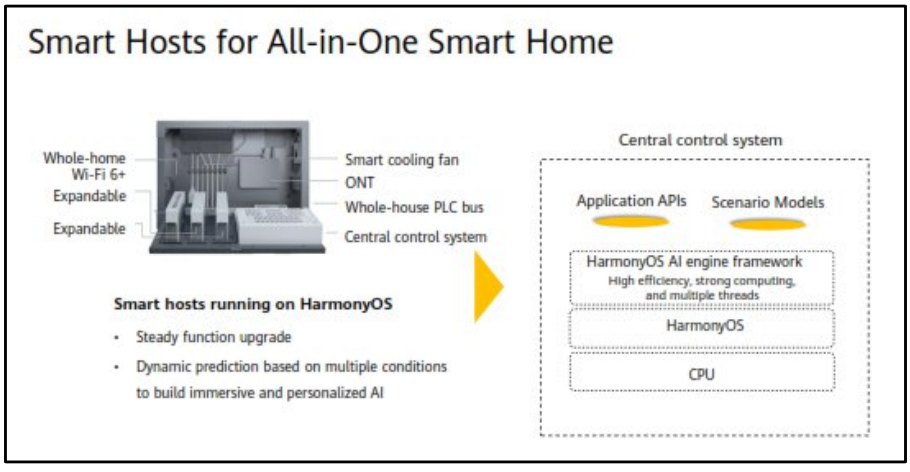
2 Solutions: 2 Interaction Modes

Two different interaction modes, providing users with flexibility in how they communicate and control smart devices, such as through voice commands, touchscreens, or mobile apps.

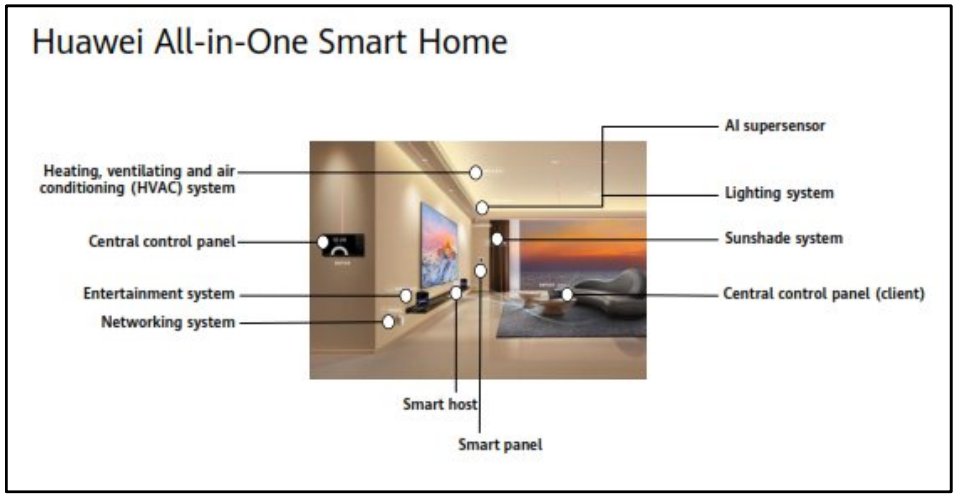
N Subsystems: HarmonyOS Connect Ecosystem

An open ecosystem accommodating various smart devices, fostering integration and offering a diverse range of products for different aspects of daily life.

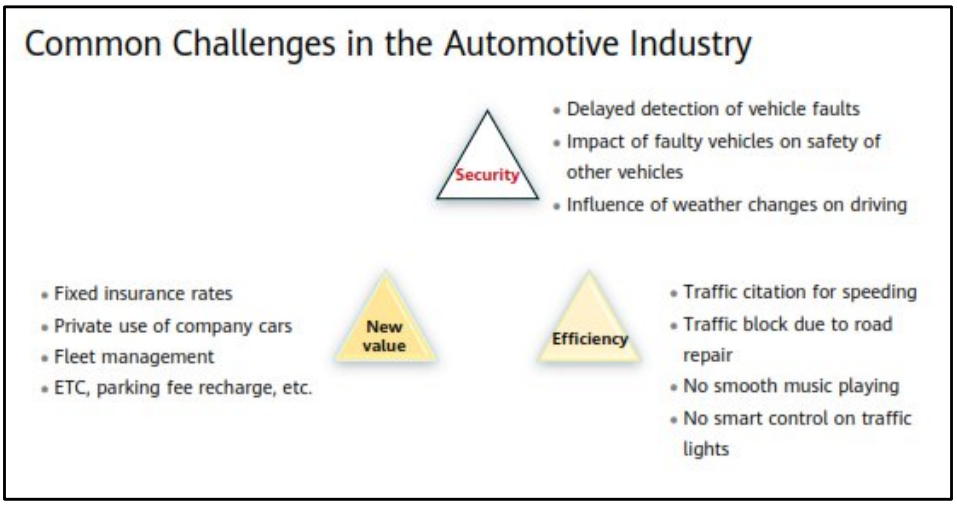
The core of the solution lies in a Harmony-powered smart host, supported by two interaction modes and an extensive range of products within the HarmonyOS Connect ecosystem. This approach emphasizes integration, adaptability, and user-friendly interactions in creating a holistic and intelligent smart home experience.



In summary, these Smart Hosts for All-in-One Smart Home are designed to be robust and adaptable, providing a comprehensive and intelligent foundation for a connected and automated smart home experience.



4. IoV Solution



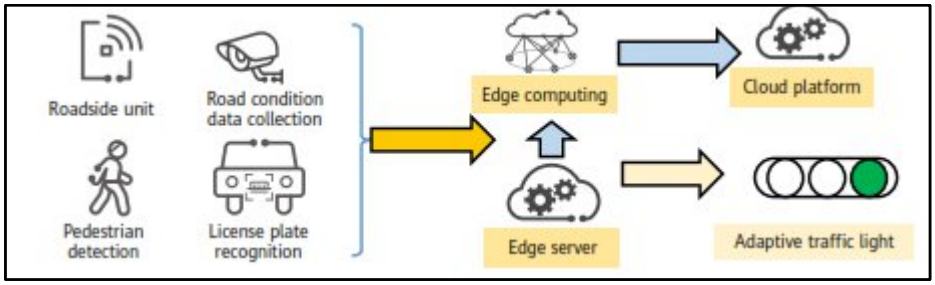
What Is IoV?

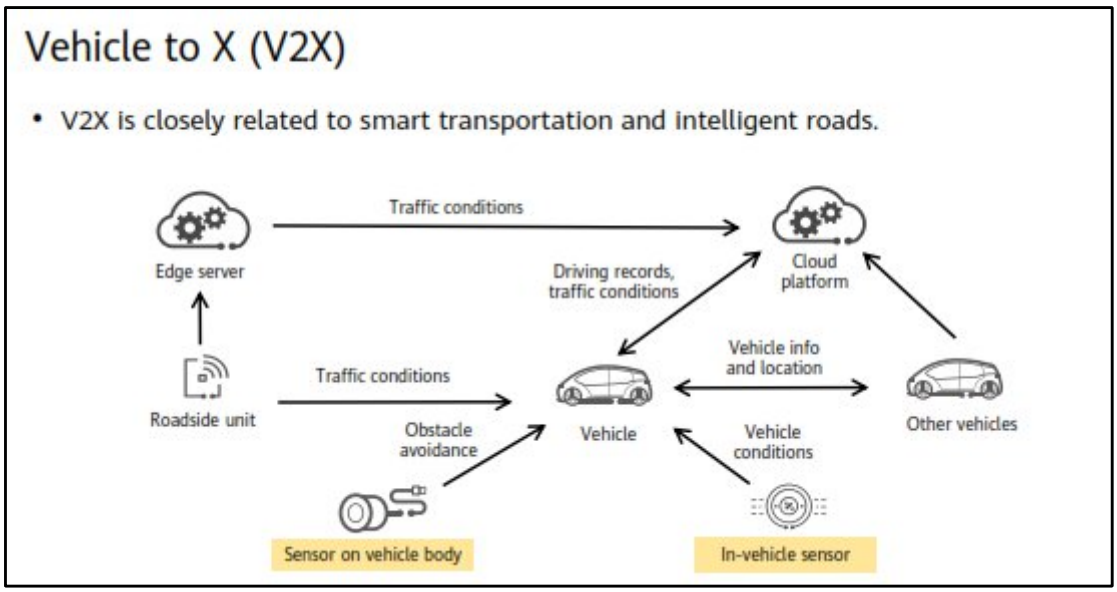
- Internet of Vehicles (IoV) means that **vehicle-mounted devices** use **wireless communication technologies** to make full use of all dynamic vehicle information on the **information network platform** and **provide various functions and services** during vehicle running.
- IoV has the **following characteristics**: Provides assurance for a distance betweenvehicles to reduce a probability of vehicle collision accidents. Provides real-time navigation for drivers and communicates with other drivers and network systems to improve traffic efficiency.

Building Smart Intersections Based on the IoV Solution

- Traffic conditions change at all hours of the day, and traffic lights are key to controlling traffic.
- Traditional solution: Traffic lights turn on and off in a specified sequence, which cannot change according to road conditions.

- Smart intersections based on IoV: Road condition data is collected to the edge to adjust the traffic lights. The cloud platform stores such data for big data analytics, providing data basis for future urban planning.





Various types of connections within the context of Internet of Vehicles (IoV):

Vehicle-To-Network (V2N):

Commonly used, allows vehicles to connect to cloud servers via a mobile network for functions like navigation, entertainment, and anti-theft.

Vehicle-To-Vehicle (V2V):

Aims to implement collision warning and congestion avoidance through information exchange between vehicles, enabling communication without speakers.

Vehicle-To-Infrastructure (V2I):

Enables vehicles to exchange data with roadside infrastructure, useful for automated driving by detecting obstacles and collecting information about the environment.

V2P (Vehicle-To-Pedestrian):

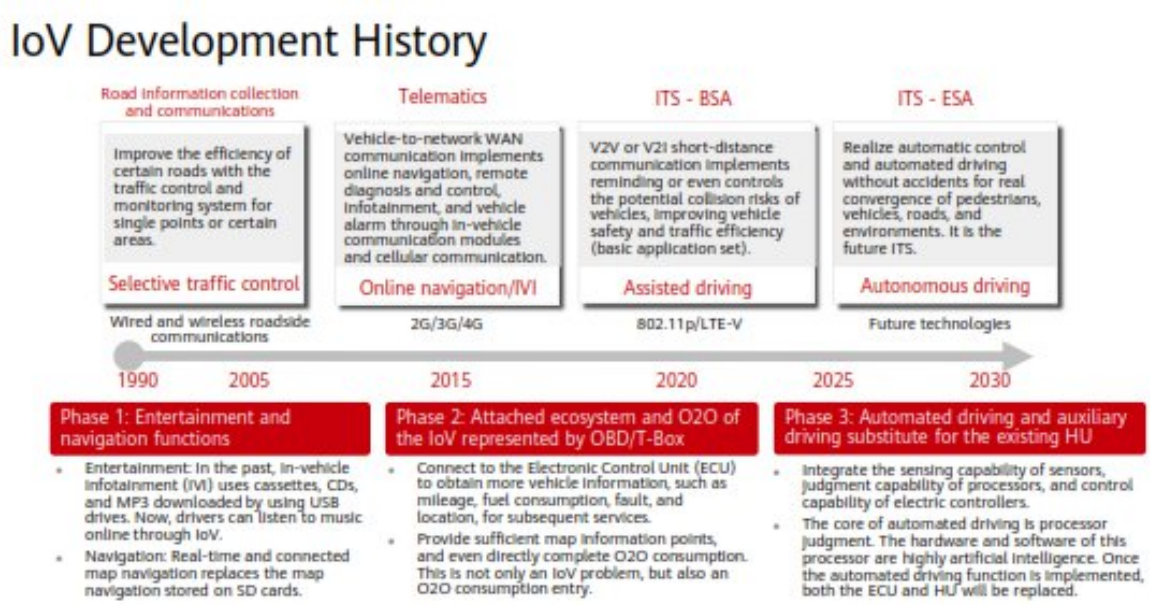
Similar to V2V, focuses on preventing collisions through information exchange, involving interactions between vehicles and pedestrians using devices like phones or wearables.

Vehicle to Everything (V2X):

As part of IoV solutions, integrates digital technologies like cloud, IoT, AI, and 5G to facilitate efficient collaboration among people, vehicles, roads, and clouds. The goal is to create smart vehicles and roads, enhancing overall performance and safety.

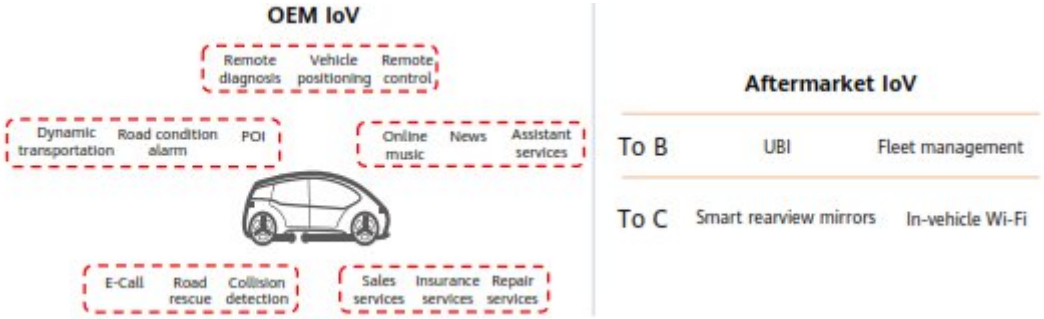
Digital Road Infrastructure Service (DRIS)

- DRIS implements digital information exchange among pedestrians, vehicles, roads, and networks, improves driving safety and traffic efficiency, and facilitates large-scale commercial use of autonomous driving.
- DRIS consists of V2X Server in the cloud and V2X Edge at the edge.
  - V2X Server provides digital road infrastructure services and edge-cloud synergy services such as data analysis and roadside computing unit management.
  - V2X Edge provides real-time service processing such as roadside sensor data collection, event identification, and communications.
- The goal of DRIS is to connect multiple roadside sensors for digital perception and provide information for efficient traffic.



Today's IoV

IoV is evolving from IVI services to smart transportation. The OEM mode focuses on internal services of OEMs, vehicle data collection, and personal entertainment information service. The aftermarket mode focuses on industry applications, supplemented by personal IVI information service.

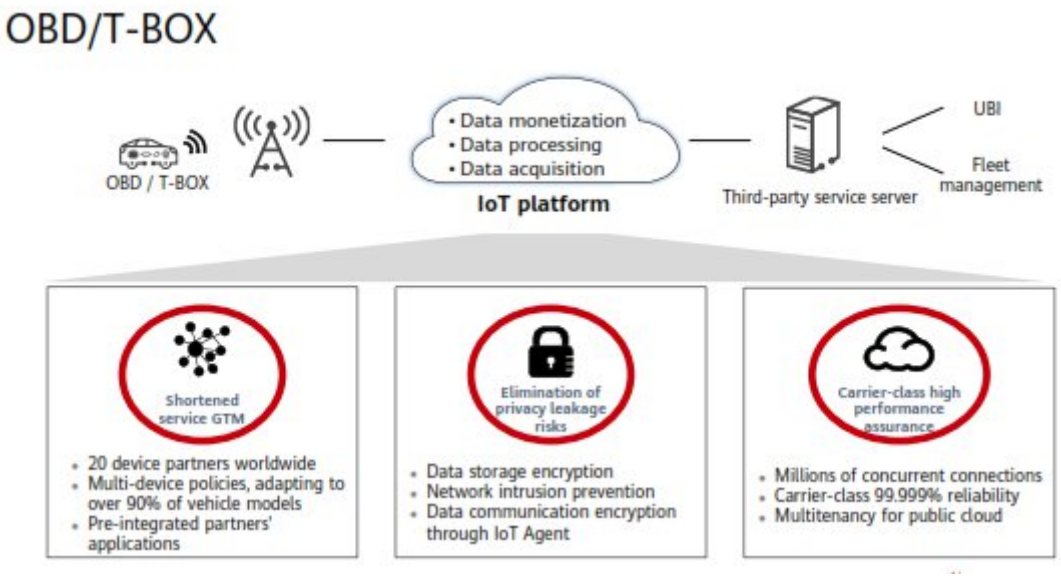


Heads Up Display (HUD)

Driving safety is the motivation for using HUD. With HUD, drivers do not need to shift the attention to the dashboard or touch the head unit during driving. The information displayed by HUD is the driving condition indicators of the vehicle, for example, speed and fuel volume. HUD also makes vehicles more intelligent by providing functions including navigation, SMS, phone, email, and simple interactions.

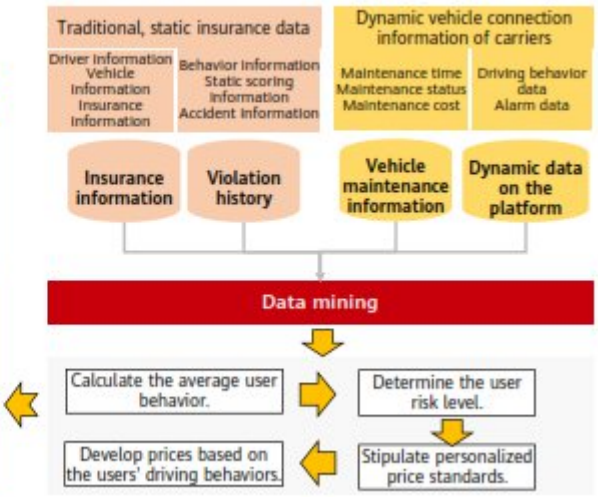
• Three HUD implementations:

- Use the front windshield as the display screen, which is a common practice in OEM mode, for example, HUD provided by automakers such as BBA.
- Use an additional display screen, which is a common practice in aftermarket mode, for example, HUD provided by Nandy, the star product for this solution.
- Use the reflection of the screen on the front windshield (or protective film), for example, AutoNavi's HUD. This implementation delivers the worst performance.

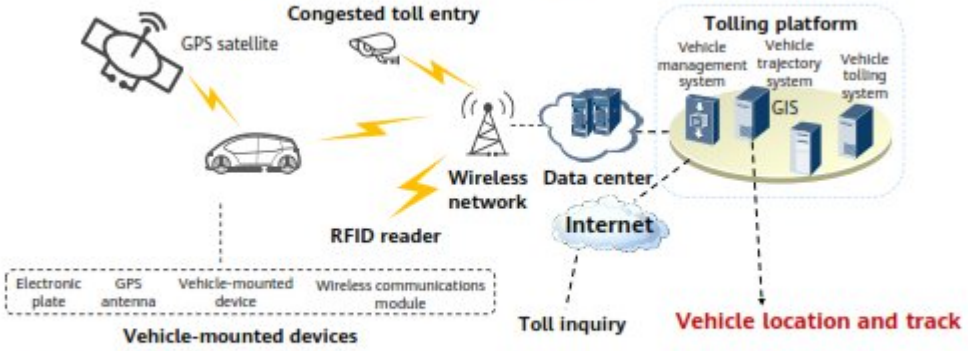


UBI

Customer Category	Business Strategy
High-risk customer	• Do not provide insurance services.
Customers with poor behavior	• Provide daily driving behavior scoring reminders and driving behavior suggestions. • Regularly communicate with the customer based on the customers' driving behavior through the call center. • Encourage customers to improve their driving behavior through discounts.
Customers with neutral behavior	• Provide daily driving behavior scoring reminders and driving behavior suggestions. • Encourage customers to improve their driving behavior through discounts.
High-value customers	• Provide more additional services to retain customers. • Perform regular surveys through the call center to increase customer loyalty.

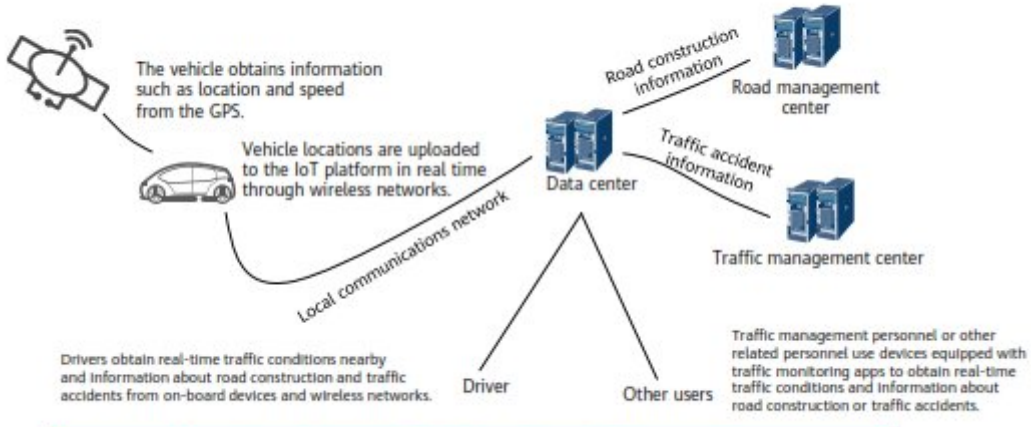


Automatic Tolling and Vehicle Trajectory Data



- A front-end video device is deployed at a congestion toll collection entry to identify license plates. RFID readers read electronic license plate information and report it to the tolling platform through the wireless network.
- Vehicle-mounted intelligent devices report the GPS data of vehicles in real time through the wireless network to determine the vehicle track.
- Supports toll collection by time or distance the congestion conditions. The tolling time and range can be set flexibly.
- Vehicle-mounted devices use GPS to determine vehicle locations, report them to the tolling platform in real time through the wireless network, and calculate the vehicle track based on GIS for automatic tolling.

Smart Transportation



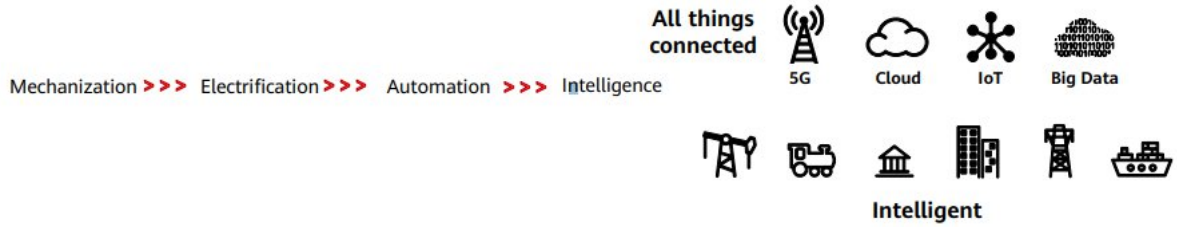
Accurate, real-time traffic data relieves traffic congestion.

5. Industrial IoT Solution

Today: The Momentum of a New Industrial Revolution

After mechanization, electrification, and automation, we have ushered in the fourth industrial revolution represented by intelligence. Intelligence is embedded into every connected things and all business processes.

ICT technologies, such as big data analytics, cloud computing, mobility, and IoT, lay the foundation of the fourth industrial revolution. All things connected Intelligent Mechanization 5G Cloud IoT Big Data Electrification Automation Intelligence



Crossing the Chasm and Developing Smart Manufacturing with Efficiency and Refinement

Focus only on the automation of production and manufacturing.

- Industry chain Information silos
- No visibility into production data

Chasm: more efficient and refined smart manufacturing (Fully connect people, data, and machines, and combine big data analytics to develop towards more efficient and refined manufacturing)

Horizontal integration of industry chain :

- Crowdsourcing and collaborative R&D
- E-commerce for marketing procurement
- IoV

Vertical integration of production information:

- Dynamic sensing and intelligent O&M
- Dynamic production control
- 3D virtual factory

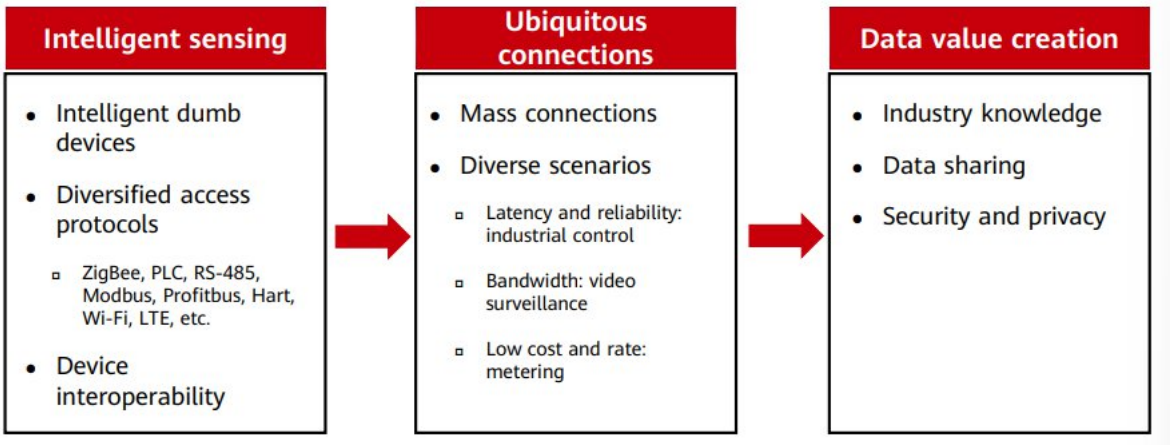
Production IoT

- Industrial cloud and intelligent big data analytics
- Mobility, real-time grasp of production information

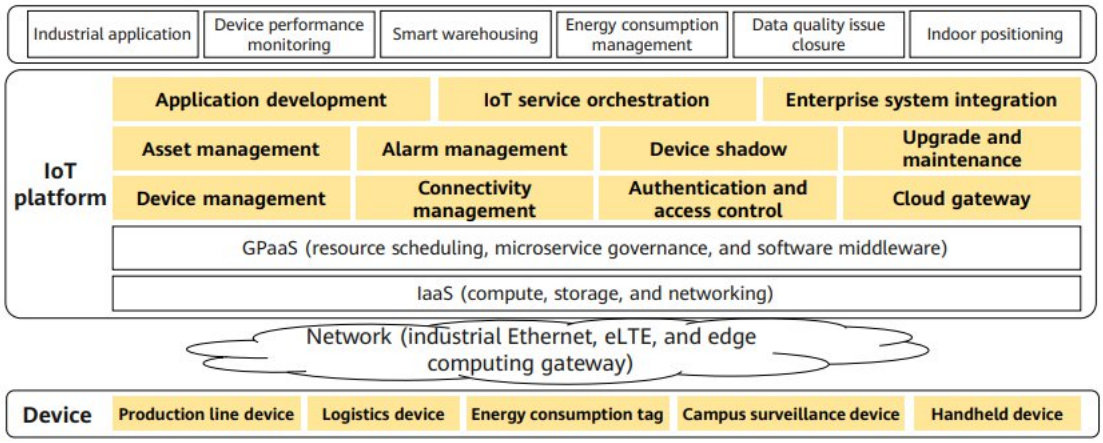
Industrial IoT Development Plans Across Countries

	2014–2017	2020–2022 Industrial IoT	2025
Germany	✓ Implemented Industry 4.0.	✓ Complete manufacturing **communications standardization**.	✓ **Unify EU Industry 4.0 standards**.
China	✓ Released *China Manufacturing 2025* initiative.	✓ Establish a **standards system** for smart manufacturing communications equipment. ✓ Chinese industrial robots occupy 50% of the market. ✓ Large-scale use of industrial wireless networks with a bandwidth of **500 Mbit/s**	✓ Breakthroughs in **10** fields, such as automotive, healthcare, and energy ✓ Large-scale use of industrial wireless communications networks with a bandwidth of **2 Gbit/s**
US	✓ Released the *Revitalize American Manufacturing and Innovation Act*.	✓ Complete flexible production line assembly within 24 hours.	✓ Invest in 1.9 billion dollars to build **45** innovation organizations. ✓ Complete flexible production line assembly within 8 hours.
Japan	✓ Released new strategies for robots and IoT. ✓ Released the Industrial Value Chain Initiative (IVI).	✓ Complete **international standardization** of manufacturing robots. ✓ Large-scale commercial use of service robots.	✓ Improve manufacturing informatization level from 30% to **50%**.
South Korea	✓ Proposed *Manufacturing Industry Innovation 3.0*.	✓ Develop IoT/smart manufacturing, make **30% existing factories** intelligent, and develop 10,000 intelligent production lines.	✓ Invest 23 billion dollars in **13** industries, such as UAVs, smart vehicles, and healthcare. The export amount exceeds Japan's.

ICT as the Production System of Smart Manufacturing Enterprises



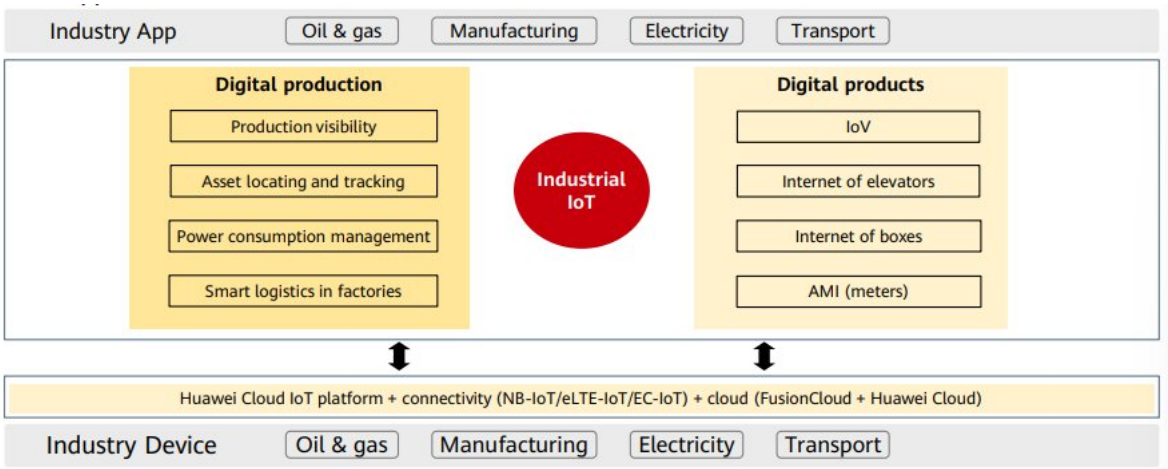
Technical Architecture of Huawei Industrial IoT Solution



As a key component of Industrial PaaS, the IoT platform provides capabilities such as device access, connectivity management, device management, and device data processing, and opens the capabilities to upper-layer applications.

Digital Production in the Factory and Digital Products Outside the Factory

Huawei provides the Industrial IoT platform, and ecosystem partners develop industry-specific devices and applications.



Necessity and Value of Visual Production Systems

Make data intuitive for production and decision-making.

Necessity :

For industry

- Improve production reliability, equipment efficiency, and management effectiveness, helping automakers stay competitive in a long term. Improve the supervision, management, and control on production processes, ensure the quality and reputation, and reduce costs. Provide valuable experience for product update, iteration, and upgrade.

For development

- Provide managers and decision makers with a strong basis to know and decide. Integrate intelligence and management, as well as automation and efficiency, to realize automated and intelligent vehicle manufacturing and more efficient personnel management.

For data application

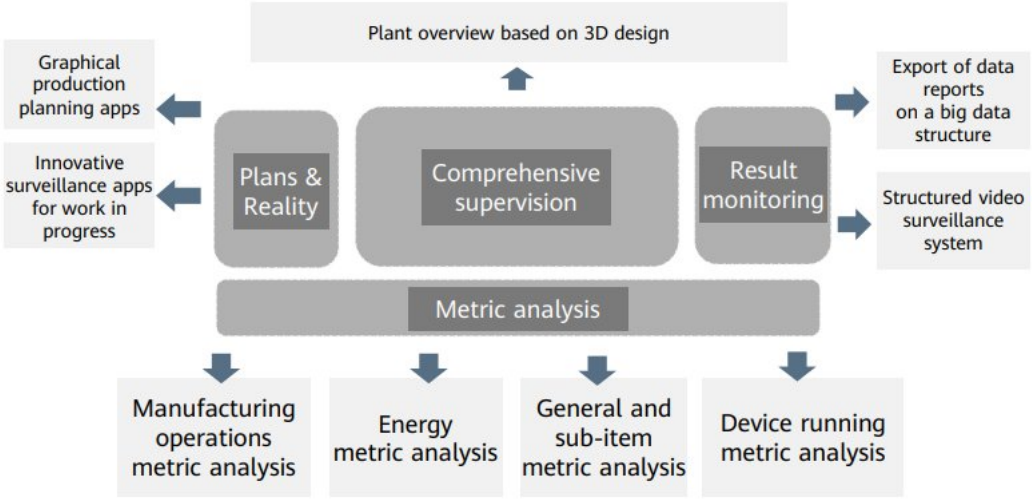
- Control asset running, predict energy consumption, and improve profitability. Make data smart and use it to make informed decisions quickly.

Value :

1. Improve efficiency and reduce costs.
2. Precisely control quality, optimize manufacturing costs, and improve profitability.
3. Display potential risks and problems during production and management in a timely manner, perform situation analysis, and release information for decision-making in time.

4. Improve the efficiency of personnel, equipment, and production lines, monitoring and management mechanisms, and production capacity control.
5. Monitor metrics in real time and ensure effective management.
6. Coordinate management and production. Use visualized, data-based, graphical, and mobile applications to eliminate potential risks and silos in production management.
7. Coordinate production factors to build robust apps and services thanks to the monitoring, management, control, and tracing capabilities of the visual systems.

Integrated Smart Factory Production Management Platform



Asset Locating and Tracking Demands

Location query

- Display the locations of people, vehicles, objects, and tools in the workshop in real time.
- Quickly find the exact location of the target and plan the arrival route by entering information on the management platform.

Bidirectional transmission of production dispatch instructions

After the AGV arrives at the station, information transmission is automatically triggered. The LED screen displays the work order and item attributes. The worker confirms the operation and returns the instruction.

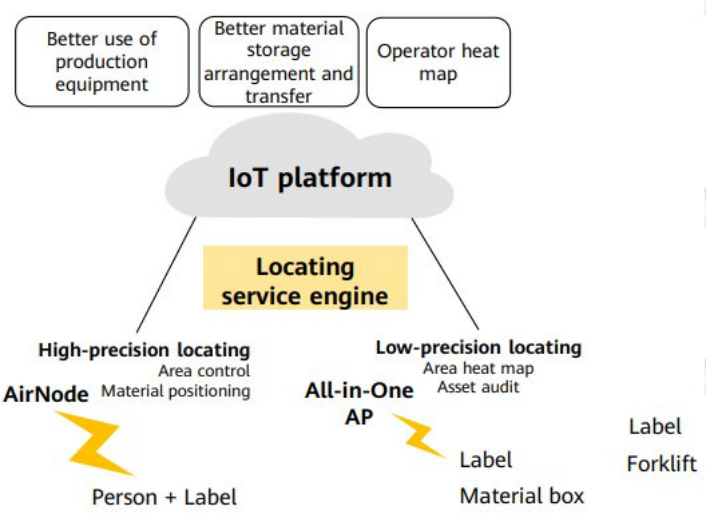
Intelligent judgment

1. Determine whether the cart or materials are at the correct location.
2. When an employee enters an undesignated Area division station, an audible and visual alarm is triggered and the system records the event. The aim is to prevent cross-work.

Area division

1. Divide areas to set alarms for each area
2. Taking away production tools and materials in specified areas will trigger alarms.
3. Unauthorized persons, objects, or vehicles will trigger alarms to the surveillance center.

Asset Locating and Tracking Solution



Platform and application systems:

The IoT platform synchronizes the location data sent by the AirNode/All-in-One AP and calculates the location using the positioning engine. Based on the output coordinates of the positioning engine, the application system matches the indoor map and provides services such as heat map analysis, foolproof operation alarm, and personnel tracing and optimization according to service needs.

Location data collection:

The key is labeling location information and periodically reporting location beacons. Labels use ultra-low power consumption chips and work continuously for more than three months. (The time changes according to the data reporting frequency.)

Network location:

The positioning base station (AirNode/All-in-One AP) receives beacons sent by positioning labels and sends the beacons to the upper-layer IoT platform for parsing positioning information. High-precision positioning accuracy: 30 cm; lowprecision positioning accuracy: 3 to 5 m